CSBP711 Advanced Artificial Intelligence — Paper Critique

Does a Vision Transformer really beat CNNs on pediatric pneumonia X-rays? A reality check of Singh et al. (2024)

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1. Summary and positioning

Primary paper. Singh, Kumar, Kumar, Verma, Abhishek & Selvarajan, *Efficient pneumonia detection using Vision Transformers on chest X-rays*, Scientific Reports 14:2487 (2024). **Problem:** classify a child's frontal chest X-ray as normal or pneumonia. **Claim:** an ImageNet-pretrained Vision Transformer (DeiT-Base/16, 224 px) detects pneumonia more accurately than CNN architectures, reaching 97.61 % accuracy (sensitivity 0.949, specificity 0.981, AUC 0.966) on 586 test images, against 76–94 % for fifteen CNNs (their Table 6). **Method:** the Kermany et al. (2018) pediatric dataset (5,856 images, Guangzhou Women and Children's Medical Center), split 80/10/10, DeiT-B fully fine-tuned with Adam, LR 1e-5 decayed ×0.995 per epoch, batch 16, 30 epochs, cross-entropy loss. **Evidence:** one training run, one random split, accuracy with a Wald confidence interval, a confusion matrix, ROC and precision–recall curves, and the CNN comparison table.

Why it matters for us. It is the same dataset and the same task as our Assignment 1, where five models under one protocol gave ResNet-50 the best AUROC (0.971) but only 86 % accuracy, after we found patient leakage, hidden duplicates and an image-geometry shortcut in the data. The paper reports 97.6 % on the same images with a model we had not tried. Either the transformer is genuinely better, or the number comes from the evaluation protocol. Our pipeline can separate the two: it can run their model under our protocol and their protocol on our models. **Supporting papers:** Dack & Dai, ICCV Workshops 2025 (dataset bias and pixel-shuffling tests on chest X-rays); Khadidos et al., arXiv 2026 (preprint: EfficientNet-B0 and DenseNet-121 on the official test folder reach 85 % and 80 % accuracy, AUC 0.97–0.98, close to our A1 numbers).

2. Critical analysis

Strengths. A clear, falsifiable claim; a public dataset; the full hyper-parameter table (their Table 4); a confusion matrix from which every metric can be recomputed; and a cost note (training was slower than the CNNs). The model choice is reasonable and, as we show below, their headline number is reproducible under their protocol.

Weakness 1 — the baselines were not fine-tuned (Table 6). The table lists trainable parameters: ResNet-50 has 200,706 trainable and 23.6 M non-trainable, DenseNet-121 100,354 trainable; the ViT has all 85.8 M trainable. The CNN backbones were frozen and only a small head was trained, while the transformer was fully fine-tuned. The text says "all hyperparameters were kept the same", but the models were not trained the same way. A 'Ref.' column next to every baseline also leaves open whether some CNN numbers were run or copied. The comparison therefore measures fine-tuning versus feature extraction, not transformers versus CNNs.

Weakness 2 — the split ignores patients and uses the training pool as test. The paper states only "80 % … training, 10 % test, 10 % validation"; it never mentions patient identity or the dataset's own test folder. In this dataset 1,155 children contribute several scans (up to 30), and in Assignment 1 we measured that a naive by-image split places 61 % of held-out images beside a scan of the same child in training. Memorising a child is then rewarded as accuracy. Separately, the official test folder is a different collection from the training folder (every A1 model scored ≥ 0.99 AUROC on validation but 0.90–0.97 on that test set); a random 10 % of the pool never faces that shift.

Weakness 3 — the numbers do not reconcile. Table 3 gives the test set as 330 pneumonia / 256 normal, but the confusion matrix (TP 152, FN 8, TN 420, FP 6) implies 160 pneumonia / 426 normal. The F-score is 0.952 in Table 5 and 0.950 in Table 6; the loss is cross-entropy in one section and binary cross-entropy in another; the stated CI (96.2–98.9 %) does not match their own Eq. 7 (96.4–98.9 %); the text says 5,863 images, the tables sum to 5,856. None of these is fatal alone, but together they show the evidence was not checked carefully.

Weakness 4 — evaluation hygiene. A single run with no seed, no cross-validation and no variance; test accuracy is tracked per epoch alongside validation ("validation accuracy gradually improves along with test accuracy"), which risks selecting on the test set; the headline metric is accuracy on a set that is 56–73 % pneumonia, where a model that always says 'pneumonia' already scores the majority share. **Limitations and reproducibility.** Limitations are one sentence ("data scarcity and the need for real-world validation"); leakage, imbalance, duplicates and the geometry shortcut are not mentioned. No code, no split files, no seeds are released, so the experiment can only be approximated. **Verdict:** the claim is larger than the numbers. The numbers support "a fine-tuned DeiT-B reaches ~98 % on a random image split"; they do not support "ViT outperforms CNNs".

3. Reality check

We added the paper's model to the Assignment 1 pipeline (timm `deit_base_patch16_224`, ImageNet weights, fully fine-tuned, LR 1e-5 as in the paper) and added a switch that reproduces the paper's protocol: pool the 5,824 de-duplicated images and draw a random, label-stratified 80/10/10 split by image. Everything else is identical to A1: grayscale, pad-to-square, 224 px, the same augmentation, AdamW, 12 epochs, checkpoint by best validation AUROC, seeds 42 and 43, one T4 GPU. Under the paper's protocol, 336 of 583 test images (58 %) share a patient with the training set; under ours, none.

Model	Protocol (test set)	AUROC	Accuracy %	Sens.	Spec.
ViT DeiT-B/16 — reported	Singh et al.: random 80/10/10 by image, n = 586, 1 run	0.966	97.6	0.949	0.981
ViT DeiT-B/16 — ours	Singh et al. protocol in our pipeline, n = 583, 2 seeds	0.997 ± 0.002	98.0 ± 0.6	0.984	0.972
ResNet-50 — ours	Singh et al. protocol in our pipeline, n = 583, 2 seeds	0.995 ± 0.004	97.2 ± 0.4	0.969	0.978
ResNet-50 — reported	Singh et al., backbone frozen (Table 6)	n/r	82.4	n/r	n/r
DenseNet-121 — reported	Singh et al., backbone frozen (Table 6)	n/r	91.8	n/r	n/r
ViT DeiT-B/16 — ours	A1 protocol: patient-grouped, official test folder, n = 618, 2 seeds	0.973 ± 0.001	82.4 ± 2.7	0.999	0.530
ResNet-50 — ours (A1)	A1 protocol, as above	0.971 ± 0.001	86.2 ± 3.4	0.997	0.636
DenseNet-121 — ours (A1)	A1 protocol, as above	0.936 ± 0.017	82.2 ± 2.7	0.999	0.526
Small CNN (scratch) — A1	A1 protocol, as above	0.942 ± 0.006	82.6 ± 0.3	0.996	0.541
Logistic regr. (pixels) — A1	A1 protocol, as above	0.895 ± 0.006	77.4 ± 4.2	0.982	0.426
Majority class — A1	A1 protocol, as above	0.500	62.6	1.000	0.000

Table 1. Reported numbers from Singh et al. versus our runs under their protocol and under the Assignment 1 protocol. ± is the std over seeds 42 and 43; n/r = not reported. Sens./Spec. = pneumonia recall / normal recall.

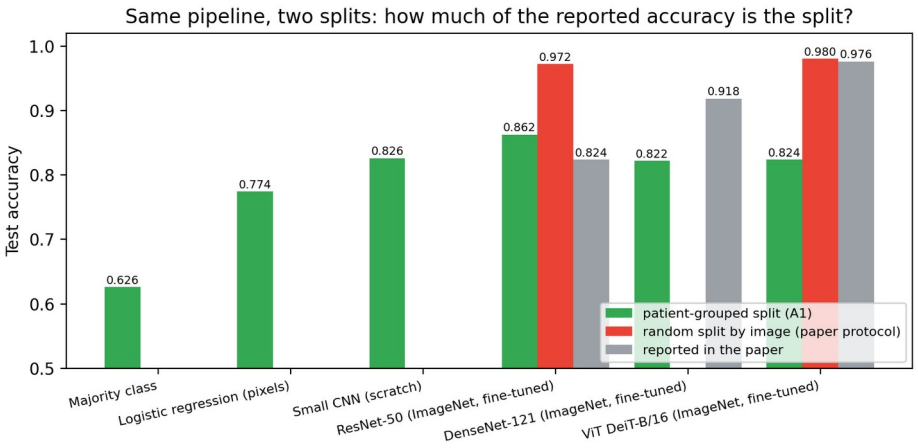


Figure 1. Test accuracy of the same pipeline under the two splits, with the paper's reported numbers in grey.

Gaps between the two setups and what each does. (a) *Split*: random by image vs patient-grouped — tested, see below. (b) *Test collection*: their test is a sample of the training pool; ours is the dataset's own test folder, a different collection — tested jointly with (a). (c) *Baseline treatment*: frozen CNN backbones vs full fine-tuning — tested (ResNet-50 row). (d) *Duplicates*: we remove 32 byte-identical copies, they do not; (e) *preprocessing*: plain resize vs pad-to-square (in A1, squashing raised ResNet-50 specificity from 0.57 to 0.81, so this gap favours them); (f) *budget*: 30 vs 12 epochs, batch 16 vs 32, Adam vs AdamW; (g) *repeats*: one run vs two seeds. Gaps (d)–(g) were not closed; their combined effect is bounded by the fact that our reproduction already exceeds their number.

Outcome. (1) **The 97.6 % reproduces, and it is the split.** Under their protocol our DeiT-B reaches 98.0 ± 0.6 % accuracy and specificity 0.97, within noise of their 97.6 % / 0.98. The same model under the patient-grouped split and the official test folder drops to 82.4 % and specificity 0.53, while its validation AUROC is 0.999. The property responsible is in the data: 58 % of test images have a sibling scan of the same child in training, and the official test folder is a different collection. (2) **The transformer does not outperform the CNN.** Under their own protocol a fully fine-tuned ResNet-50 scores 97.2 ± 0.4 %, one point from the ViT and 15 points above the 82.4 % they report for ResNet-50 with a frozen backbone (gap c). (3) **Under fair conditions the ViT ties on AUROC and loses on cost:** 0.973 vs 0.971 (seed spread 0.001), with 3.7× the parameters (85.8 M vs 23.5 M), 2× the training time (9.2 vs 4.4 min) and lower accuracy and

specificity. The paper's claim does not hold up; what holds up is "evaluation protocol dominates architecture on this dataset".

4. Next steps

Close the remaining gaps in one run each: their 30-epoch budget and plain resize under the patient split, to confirm they do not change the ranking; a third seed. Then test the transformer where its inductive bias could matter: an external pediatric test set (e.g. VinDr-PCXR) rather than this dataset's own folder, and a train-size sweep, since ViTs are known to need more data than CNNs. For the course project, the same protocol audit (grouping, collection shift, baseline parity) becomes a checklist we apply to any benchmark claim before building on it.

5. AI-use statement

Claude (Anthropic) was used, under our direction, for the literature search that shortlisted candidate papers, for extracting the paper's tables, for writing the new code (the ViT model entry, the image-level split option, the A2 report module and run script) and for drafting this report and the slides. All training runs were executed by us on Google Colab; every number here comes from our own result files in the repository. The first image-split run was found to be invalid when we inspected its outputs (the split had pooled the dropped duplicate rows of the index); the code was fixed, a guard added, and the four runs repeated. We checked the paper's figures against the open-access PDF ourselves. **Contributions:** M. Hindi — pipeline changes, Colab runs, reality check and report; K. AlHassani — paper summary and critique, slides; A. AlKaabi — gap analysis, figure, repository and README. [adjust before submission]

References

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